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Original Article

IMU Calibration Effect on Lower Limbs Kinematics Against Optical Motion Capture in Post-Stroke Gait

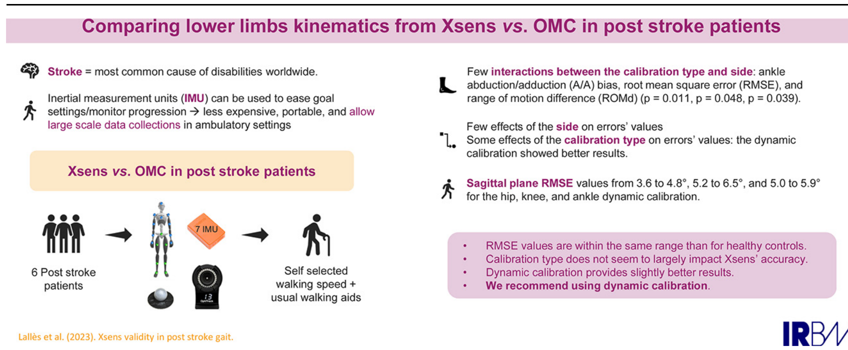
Ariane P. Lallès [a,b,*], Geoffroy Moucheboeuf [c,d], Emilie Doat [e], Hélène Pillet [a], Xavier Bonnet [a]

[a] ENSAM, Institut de Biomécanique Humaine G. Charpark, 151, Boulevard de l'Hôpital, F-75013 Paris, France
[b] Department of Sport sciences and physical education, Ecole normale supérieure de rennes, Campus de Ker Lann, Avenue Robert Schumann, F-35170 Bruz, France
[c] Service de Médecine Physique et Réadaptation, Pôle de neurosciences cliniques, Centre Hospitalier Universitaire de Bordeaux, Bordeaux, France
[d] HACS team-U1219 INSERM Bordeaux Population Health & University of Bordeaux, Bordeaux, France
[e] Université de Bordeaux, CNRS, INCIA, UMR 5287, F-33000 Bordeaux, France

HIGHLIGHTS

- Errors between motion capture and IMUs for post-stroke match literature ranges.
- Calibration type and slow speed does not impact Xsens' accuracy to a great extent.
- Dynamic calibration provides slightly better results.
- We recommend using dynamic calibration based on the patient's walking ability.

GRAPHICAL ABSTRACT



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ABSTRACT

Background: Stroke is the most common cause of disabilities worldwide. Rehabilitation is central to restore functions. Inertial measurement units (IMU) can be used to ease goal settings and monitor progression. Contrary to optical motion capture (OMC), IMU are less expensive, portable, and allow large scale data collections in ambulatory settings. Although Xsens MVN system validity has been demonstrated in healthy participants, its validity among post-stroke (PS) patients is yet to be proven.
Research question: Computation methods being affected by the calibration type; the goal of this study is to compare lower limbs kinematics from Xsens system, after two calibrations against OMC in slow PS walkers exhibiting reduced ranges of movements.
Methods: Data was collected for six PS patients. They were equipped with 29 reflective markers and seven IMU. A minimum of two walks with a dynamic calibration and four walks with a static calibration were performed. All trials were accomplished at a self-selected walking speed and PS used their usual walking aids.
Results: Few interactions between the calibration type and side were found for the ankle abduction/adduction (A/A) bias, root mean square error (RMSE), and range of motion difference (ROMd) ($p = 0.011$, $p = 0.048$, $p = 0.039$). Few effects of the side on errors' values were found. We noticed some effects of the calibration type on errors' values, the dynamic calibration showing better results. In the sagittal plane, we reported RMSE values from 3.6 to 4.8°, 5.2 to 6.5°, and 5.0 to 5.9° for the hip, knee, and ankle dynamic calibration.

* Corresponding author at: ENSAM, Institut de Biomécanique Humaine G. Charpark, 151, Boulevard de l'Hôpital, F-75013 Paris, France.
E-mail addresses: Ariane.lalles@ensam.eu (A.P. Lallès), geoffroy.moucheboeuf@gmail.com (G. Moucheboeuf), emilie.doat@u-bordeaux.fr (E. Doat), Helene.pillet@ensam.eu (H. Pillet), xavier.bonnet@ensam.eu (X. Bonnet).

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Significance: The calibration type, reduced range of movement, and slow walking speed does not seem to impact Xsens' accuracy to a great extent. Nevertheless, dynamic calibration provides slightly better results. Considering the patient's walking ability, we recommend using this calibration.

1. Introduction

Stroke is the most common cause of disabilities worldwide [1]. Rehabilitation's main aim is to restore function and motor recovery [2,3]. This process should start early in the acute stroke unit [2], from the acute (<14 days post-stroke (PS)) to the subacute (14 days - 6 months PS) and until the chronic (>6 months PS) phase [4].

PS gait is commonly assessed through clinical scales [5,6] that are known to have a poor reliability [7]. Quantitative methods can help to understand underlying biomechanical mechanisms [5]. Spatiotemporal and kinematic measurements have been shown to be clinically relevant markers of impaired walking performance in PS patients, highlighting their need to be accurately assessed [8]. Spatiotemporal parameters are the most used as they require less time and equipment to be assessed. Nevertheless, they do not provide any information on the underlying mechanisms of gait changes [5]. PS patients are known to exhibit reduced joint angles compared to healthy populations. Slower PS also show abnormal lower limb joint angles patterns [9]. Few authors tried to define key gait parameters in PS [5,9–12]. Multi-modal approaches revealed that key patterns were centered around gait speed, and lower limbs sagittal angles (knee flexion/extension being the most reported one). While joint kinematics could serve as a useful indicator of recovery during therapy, they remain underreported due to measurement difficulties [13,14].

Optical motion capture (OMC) is widely used as the gold standard for gait analysis. Nevertheless, these systems are expensive, require space, and offer a low ecological validity [15–17], compromising their accessibility in clinical settings. Thus, inertial measurement units (IMU) offer a good alternative as they are less expensive, portable, and allow large scale evaluations in real life environments [15,16,18]. Unlike the traditional OMC, IMU do not require high training and expertise, in both data collection and post-processing [15,19,20]. IMUs however suffers from the main drawback that they do not directly measure positions of markers like OMC. To retrieve the position and orientation of the bone segment, a specific biomechanical model and algorithms are necessary. One of the most powerful available on the market is the one developed by Xsens Technologies (Enschede, Netherlands). Several authors demonstrated Xsens validity in measuring healthy gait spatiotemporal parameters and kinematics [16,21–24]. This system has the advantage of offering a user-friendly interface for analyzing and displaying data. For example, similarities between Xsens and OMC curves are shown to be excellent in the sagittal plane for hip, knee, and ankle angles [24]. As reported by diverse authors, Xsens MVN can estimate sagittal lower limbs joint angles with a root mean square error (RMSE) close to five degrees [21–23]. This can be considered acceptable for clinical settings [25]. Most errors are attributable to differences in biomechanical models (*i.e.*, Xsens vs. OMC), and not to the IMU itself [26]. A consistent offset has been reported by several authors in the hip sagittal plane, the knee and ankle transversal and frontal planes [21,23,24]. These estimations are independent of the operator's expertise [21], making IMU accessible to clinicians. This system is usable for clinical assessments, however, Xsens proprietary algorithms still need to be assessed in pathological populations [21].

In clinical settings, wearable sensors offer a promising solution in providing valuable insights by enabling comparison of different rehabilitation training responses [13]. IMU can be used to ease goal settings, monitor progression, and thereby play a key role in refining treatment approaches and updating clinical classifications of recovery [13,18,27]. Nevertheless, a routine use requires these tools to provide valid and reliable measurements. Understanding their validity is critical to determining their potential applications, yet this validation process is often insufficiently addressed [28]. It is important to note that IMU measurement accuracy is sensitive to external perturbation of the magnetic field, which can be difficult to control in a clinical environment. IMUs could also be affected by various factors, such as reduced walking speed and kinetic changes, which may distort acceleration and rotational signals. These inaccuracies can lead to errors in detecting gait cycles and challenge the robustness of algorithms, especially when dealing with asymmetric gait patterns commonly seen in specific pathological populations [28]. Validating and optimizing the use of IMUs in rehabilitation can significantly enhance patient motivation and adherence to therapies by providing accurate feedback on progress [29]. These tools can help guide patients in properly executing their therapy, ensuring more effective rehabilitation outcomes [29]. Benefits may also extend beyond patients to caregivers and the medical team, as reliable data from IMUs can improve communication, monitoring, and treatment decisions, leading to better overall care [29].

Recently, Schwarz et al. [19] investigated Xsens MVN reliability in 28 chronic PS patients during ten meters and six minutes walking tests. Only eight PS had pronounced ambulation deficits, and most participants had a walking speed greater than 1.1 $m.s^{-1}$. Although the system provided relatively good-to-excellent test-retest reliability, no validation against the gold standard (*i.e.*, OMC) was performed. This validation process is important as PS exhibit an altered gait pattern, with reduced range of movements (ROM) and walking speed.

To our knowledge, kinematics provided by Xsens proprietary algorithms have not been validated in a pathological population. One group provided a specific protocol for assessing kinematics in cerebral palsy children, but calibration and kinematics were computed using their own model [30]. To be used in the clinic, it is important to validate Xsens outputs as clinicians will not develop their own calibration and kinematic computations.

To provide a cost- and time-effective method to quantitatively assess gait characteristics in subacute PS patients, a first step would be to validate the Xsens MVN system. This system enables two calibration types: a dynamic, involving walking, and a static, using a neutral pose (N-pose). A comparison between two static poses that are available on the software has shown higher accuracy for the T-pose (*i.e.*, neutral pose with arms straight, stretching sideways horizontally) instead of the N-pose [31]. While Xsens advise running the dynamic calibration to ensure a better biomechanical model calibration, PS patients may encounter difficulties in performing it (*i.e.*, slow walking speed, reduced ROM). This calibration method would also induce fatigue that may result in shorter data collections. Using a static calibration could overcome these limitations but would also be limited by the patients' ability to maintain a neutral pose.

Computation methods being affected by the model and the calibration type, it will be necessary to define the optimal method that enables accurate results while being achievable by the pathological population. Thereby, the aim of this study was to compare lower limbs kinematics from Xsens MVN system, after two calibrations (*i.e.*, dynamic, static), against OMC in slow PS walkers exhibiting reduced ROM.

2. Materials and methods

2.1. Participants

Participants consented to participate in this institutional review board-approved study 2020-A01357-32 at "*Centre Hospitalier Universitaire Bordeaux*" (France). All PS were recruited in the "post-acute care and rehabilitation" unit. We included adult patients that were able to understand simple orders, instructions and give an informed consent. Participants had to be able to walk 10 meters without any human physical assistance and were asked to keep their walking aids. All PS that were pregnant, couldn't walk without discomfort, pain or exaggerated fatigue were excluded. We also excluded participants that had an unstable medical condition, and an orthopedic or neurological condition (prior to the stroke) affecting their gait.

Participants' functional autonomy was evaluated using the Barthel index [32]. Ten activities of daily living are assessed (*i.e.*, bowels, bladder, grooming, toilet use, feeding, transfers, walking, dressing, climbing stairs, bathing). Scoring is based on "time and amount of actual physical assistance required if a patient is unable to perform the activity" [32]. We used the refined 0-10 points for every activity [33]. The lower the score, the higher the incapacity.

2.2. Motion analysis

Motion data were collected during overground walking using 24 optoelectronic cameras (Flex 13, NaturalPoint, Inc. DBA OptiTrack, Corvallis, OR, USA) at 120 Hz. Twenty-nine reflective markers were placed on the participant (Fig. 1).

Seven MTw Awinda trackers (XSens Technologies B.V., Enschede, Netherlands) recorded gait at 100 Hz using Xsens MVN Record 2023.0.0. Following Xsens recommendations, IMU were placed on the lower limbs (*i.e.*, foot, shank, thigh, and pelvis) using straps (Fig. 1). Two different calibrations were performed. The dynamic calibration included a static pose followed by 15 seconds of walking. The static calibration consisted in only performing the static pose. After calibration, Xsens advises to walk around to train algorithms. We did not perform this step as this would have induced fatigue and limit data collection. Subjects were instructed to walk at their usual pace, with their walking aids. To limit the risk of fall, a physical therapist intern walked next to the patient. A minimum of two trials involving a dynamic calibration were collected. This number was set to two as some PS patients were not able to perform more walks using this calibration without inducing too much fatigue. A minimum of four trials involving a static calibration were collected. This number was higher because static calibration enabled more acquisitions before fatigue. This data was also likely not to be fully usable so taking four trials enabled to have at least two good acquisitions.

The two systems were synchronized using a trigger sent from the Motive software (NaturalPoint, Inc. DBA OptiTrack, Corvallis, OR, USA) to Xsens MVN software.

2.3. Data processing and analysis

OMC joint angles were calculated in MATLAB (R2022a, MathWorks, Natick, MA, USA) using the ISB recommendations. Hip joint centers were calculated using Bell et al. [34] method. Knee joint centers were computed as the middle of the condyles, and the ankle as the middle of the malleoli. Kinematics were filtered using a fourth-order low-pass Butterworth filter (cut-off frequency of 5 Hz), and time-normalized to 100% gait cycle (%GC) using Xsens foot contacts' detection. Xsens MVN lower limbs joint angles were also extracted. Gait speed was computed using the middle of OMC pelvis markers.

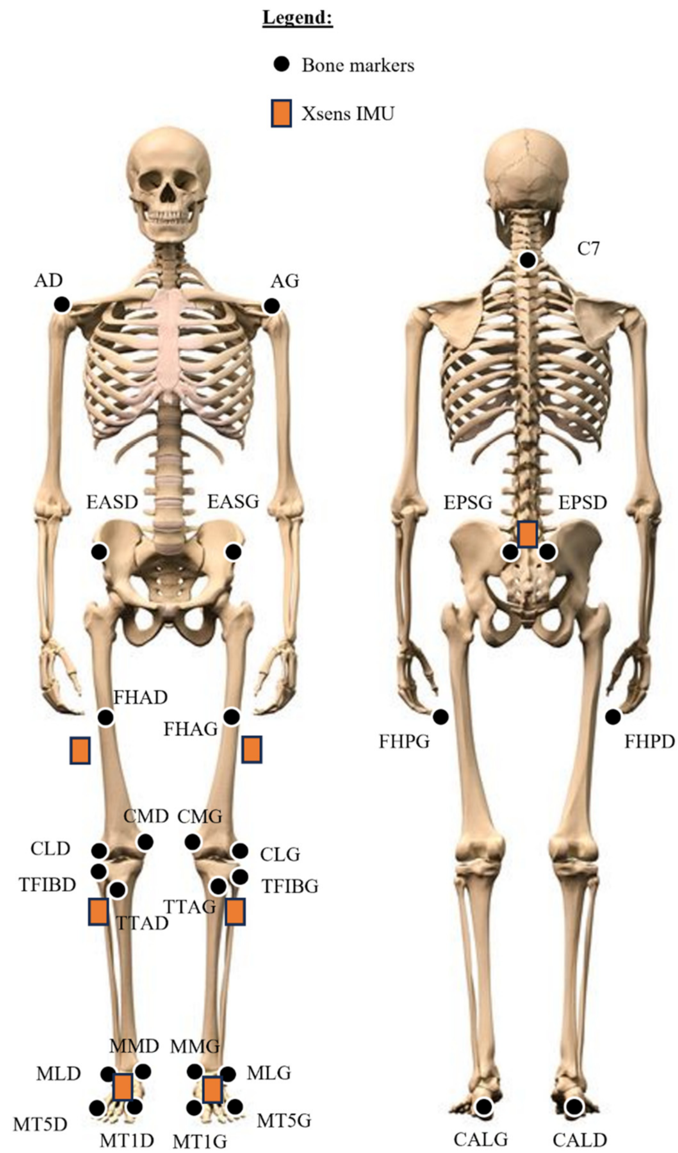

Fig. 1. Markers and IMUs positioning on the human body.

Three cycles per trial were selected and averaged. Two trials per subject and per calibration condition (*i.e.*, dynamic, static) were selected, and averaged.

For all angles, we calculated the difference between the OMC and the Xsens system. To quantify the systematic and random errors, we also computed bias and typical error of measurement (TEM) (Formula n°1). Root mean square error (RMSE) and range of motion differences (ROMd) were also computed. This was performed in all axes of rotations: flexion/ extension (F/E), abduction/adduction (A/A), internal/external rotation (I/E).

$$Bias = mean(OMC - Xsens) \quad and \quad TEM = SD(OMC - Xsens) \tag{1}$$

2.4. Statistical analysis

Statistical analyses were performed on MATLAB and R studio (R Core Team, Vienna, Austria). Normality and homogeneity of variances were assessed respectively by Shapiro Wilk and Levene's tests. A two-way ANOVA was performed to assess the effect of

Table 1
Anthropometric data for the six post-stroke patients.

N°	Sex	Age	Height (m)	Weight (kg)	BMI ($kg.m^2$)	Aff side	Hemiplegia	Time PS (days)	Barthel index	Walking aid	Gait speed ($m.s^{-1}$)
S02	M	61	1.71	104	36	L	Yes	183	65	tripod cane	0.32 ± 0.02
S03	F	54	1.56	58	24	R	Yes	254	60	crutch	0.28 ± 0.00
S04	M	57	1.78	139	44	R	-	26	80	-	0.75 ± 0.03
S05	M	68	1.60	82	32	L	-	25	75	walker	0.40 ± 0.07
S06	F	53	1.71	69	24	R*	Yes	58	75	-	0.36 ± 0.01
S07	F	43	1.63	67	25	L	Yes	92	60	tripod cane	0.17 ± 0.01
Mean	-	56	1.67	87	31	-	-	106	69	-	0.38
SD	-	6	0.09	32	9	-	-	104	8	-	0.20

Aff: affected, BMI: body mass index, F: female, L: left, M: male, PS: post-stroke, R: right
*patient that underwent two strokes.

calibration type and side (*i.e.*, affected, unaffected) on the errors' values. Significance level was set to 0.05.

All parameters are presented with mean ± standard deviation (SD). All curves are presented with mean ± standard error (SE).

3. Results

Data were collected for six patients whose characteristics are presented in Table 1. Four participants were in their subacute PS phase. Most of participants required a walking assistance such as canes or crutches. Only one participant had a walking speed greater than 0.4 $m.s^{-1}$.

3.1. Angle measurements

Joint angles curves are presented in Figs. 2 to 4. For the affected hip F/E, we can notice a difference in the maximum extension between OMC and Xsens, especially for the dynamic calibration (Fig. 2). Xsens system provided a lower extension value compared to OMC. At the knee level, we can clearly identify the reduced maximum flexion on the affected side (*i.e.*, <30°) compared to the non-affected one (>40°) (Fig. 3). We can also notice that Xsens did not detect any knee hyperextension during the stance phase on the affected side. There is an offset in F/E between systems on the affected side for both calibration types. Lastly, there seem to be an offset for the ankle F/E for both sides and calibrations (Fig. 4).

Range of movement (ROM) values obtained with the dynamic calibration are presented in Table 2. We can clearly identify a reduced F/E ROM on the affected side in comparison with the unaffected one for the knee (*e.g.*, IMU, 28.9 ± 11.3 *vs*. 45.1 ± 7.1°) and ankle (*e.g.*, IMU, 17.9 ± 6.1 *vs*. 26.4 ± 8.0°).

3.2. Errors between systems

Errors between systems are presented in Table 3. Negative values imply an overestimation of the Xsens system compared to the OMC. As highlighted in green, most RMSE are below 5°.

The two-way ANOVA revealed that there was statistically significant interactions between the effects of calibration and side for ankle A/A bias ($F(1, 20) = 7.84$, $p = 0.011$), ankle A/A RMSE ($F(1, 20) = 4.44$, $p = 0.048$), ankle A/A ROMd ($F(1, 20) = 4.90$, $p = 0.039$), and ankle F/E ROMd ($F(1, 20) = 6.85$, $p = 0.017$).

Simple main effects analyses showed that calibration did have statistically significant effects on TEM values for knee A/A ($p = 0.009$), knee F/E ($p = 0.003$), ankle A/A ($p = 0.019$), and ankle F/E ($p = 0.008$). We also found statistically significant effects on RMSE values for knee A/A ($p = 0.012$) and ankle A/A ($p = 0.024$). Lastly, there were significant effects on ROMd values for knee A/A ($p = 0.002$), knee F/E ($p = 0.001$), and ankle F/E ($p = 0.048$). In other words, the dynamic calibration provided lower TEM, RMSE, and ROMd values for the dynamic calibration compared to the static one.

Simple main effects analyses showed that side (*i.e.*, affected, unaffected) did have statistically significant effects on knee I/E bias ($p = 0.004$) and knee A/A RMSE ($p = 0.021$). In other words, the affected side led to lower RMSE values compared to the unaffected side.

4. Discussion

The goal of this study was to validate lower limbs kinematics from Xsens MVN system, after two calibrations (*i.e.*, dynamic, static), against OMC in slow PS walkers exhibiting reduced ROM. We found few interactions between the calibration type and side for errors values (*i.e.*, ankle A/A bias, RMSE, and ROMd). We also reported few effects of the side on errors' values. We noticed some significant effects of the calibration type on errors' values. In this context, dynamic calibration provided more accurate results compared to the static one, especially for A/A and F/E.

Up to now, no author has explicitly reported systematic and random errors for Xsens system. We found highest systematic errors for the sagittal plane. This may have been induced by the calibration pose. Indeed, Xsens MVN assumes a known body pose during calibration [23]. During the neutral pose, participants are expected to have parallel feet, pointing forward, and spaced from one foot. They also need to have the legs pointing straight down. While this seems easily achievable by healthy participants, PS patients experienced some difficulties. Some of them had difficulties in having parallel feet, and fully extended knees. Participants that had walking aids were also not fully aligned, especially at the hip. Indeed, PS patients usually shift their weight onto the non-affected side. Then, PS experienced some difficulties in during the 15 seconds walking bout as acquisition time was too short to perform enough gait cycles. This may have also induced some fatigue. After calibration's processing, subjects needed to recover the neutral pose, which was also difficult for PS patients. Besides, we did not train the algorithms by walking around as it was recommended by Xsens. Thus, body segments were not perfectly aligned with the reference pose, and may have resulted in wrong functional axes definitions, and thereby higher errors [35]. This might have been enhanced by the small ROM, especially for the knee and hip F/E on the affected side. Making a parallel with studies focusing in cerebral palsy children, the dynamic calibration could be replaced by passive joint F/E to better define functional axes [30].

As expected, RMSE values were lower for the dynamic compared to the static calibration. This is in agreement with Xsens' recommendations, and suggest, indeed, a better biomechanical model calibration. Our results are mostly in line with what have previously been reported [22,23]. Indeed, we reported RMSE values ranging from 3.6 to 4.8° for hip, 5.2 to 6.5° for knee, and 5.0 to 5.9° for ankle dynamic calibration. For A/A, we reported values

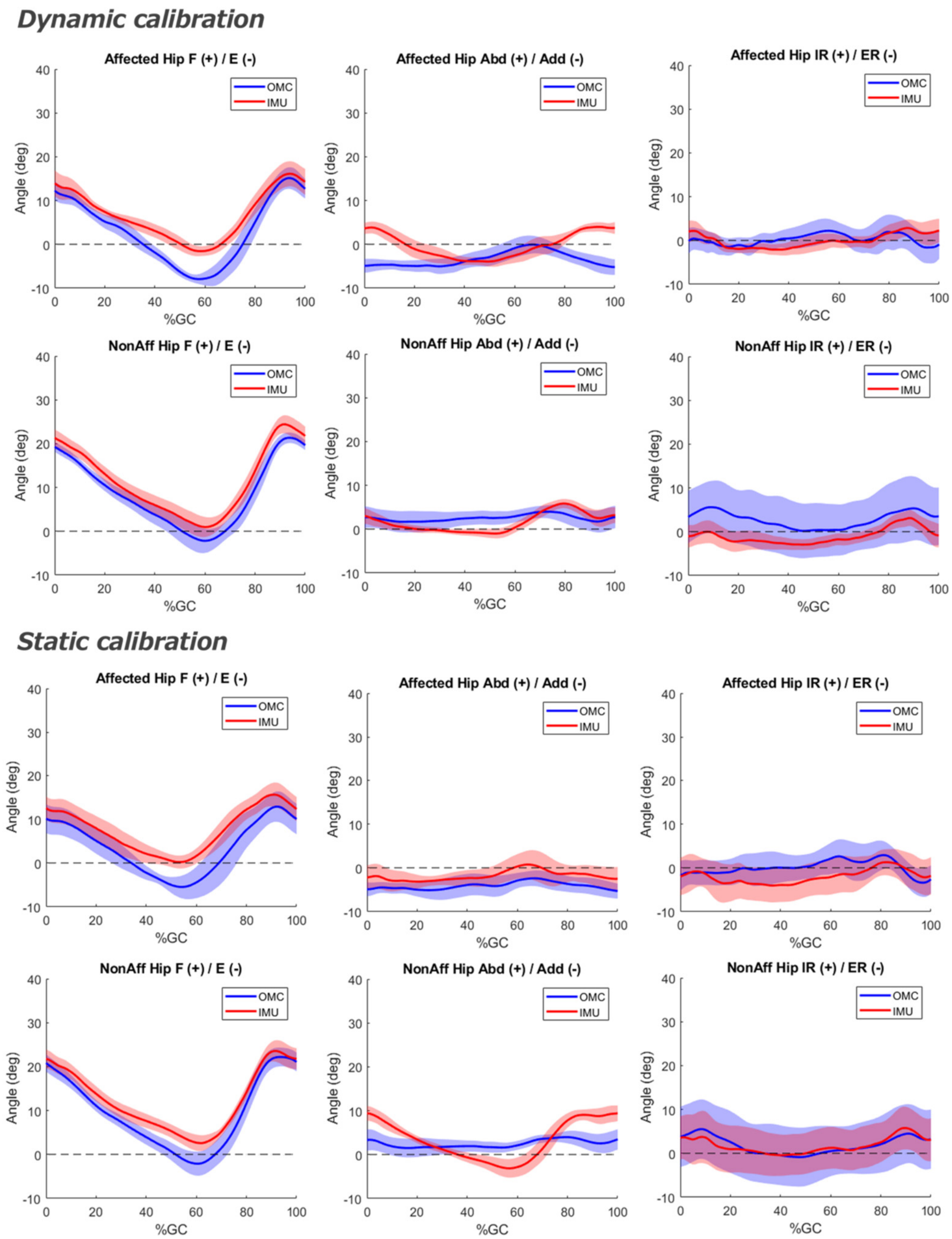

Fig. 2. Hip angles for the dynamic and static calibrations. Abd/Add: abduction/adduction, F/E: flexion/extension, IMU: inertial measurement unit, IR/ER: internal/external rotation, NonAff: non affected, OMC: optical motion capture.

ranging from 4.4 to 5.5° for hip, 1.9 to 3.5° for knee, 4.3 to 4.5° for ankle. Lastly, in the transversal plane, we reported RMSE values ranging from 3.9 to 9.6° for hip, 3.1 to 3.7° for knee, and 3.7 to 11.2° for ankle.

We also found lower ROMd values for the dynamic compared to the static calibration, especially for knee and ankle F/E. Our values are in agreement with what have been previously reported in the literature for healthy subjects [24]. However, we noted higher values for the knee I/E and F/E. This may be due to inappropriate functional axis definition as PS gait is characterized by asymmetries and lower ROM. These authors [24] used the same Xsens dynamic calibration. Therefore, we suppose that most differences in ROMd values come from the OMC. In fact, our marker set and biomechanical models were different from Zhang et al. [24]. Thus, our axes of rotations and anatomical frames were likely different. These differences are known to contribute to disparities between systems [24]. In other words, discrepancies with literature do not solely come from Xsens' kinematics, but also from OMC kinematics. One must also note that soft tissue artefacts and markers misplacement induced errors in OMC measurements [36]. Small non sagittal angles of distal joints (*i.e.*, knee, ankle) are also known to be the least precise and reproducible [22,35,37,38]. Angles with small values are also likely not accurately measured by the Xsens system [22].

To be used routinely with PS patients, we must ensure that the instrument's errors are smaller than the minimal detectable change (MDC) or minimal clinical important difference (MCID). MDC describes the minimal change that falls outside the measurement errors [39], and MCID the smallest amount of an outcome that must change to be meaningful to patients [40]. In PS patients, Guzik et al. [41] reported MCID values for the knee sagittal ROM for both the affected and unaffected sides (respectively 8.48° and 6.81°).

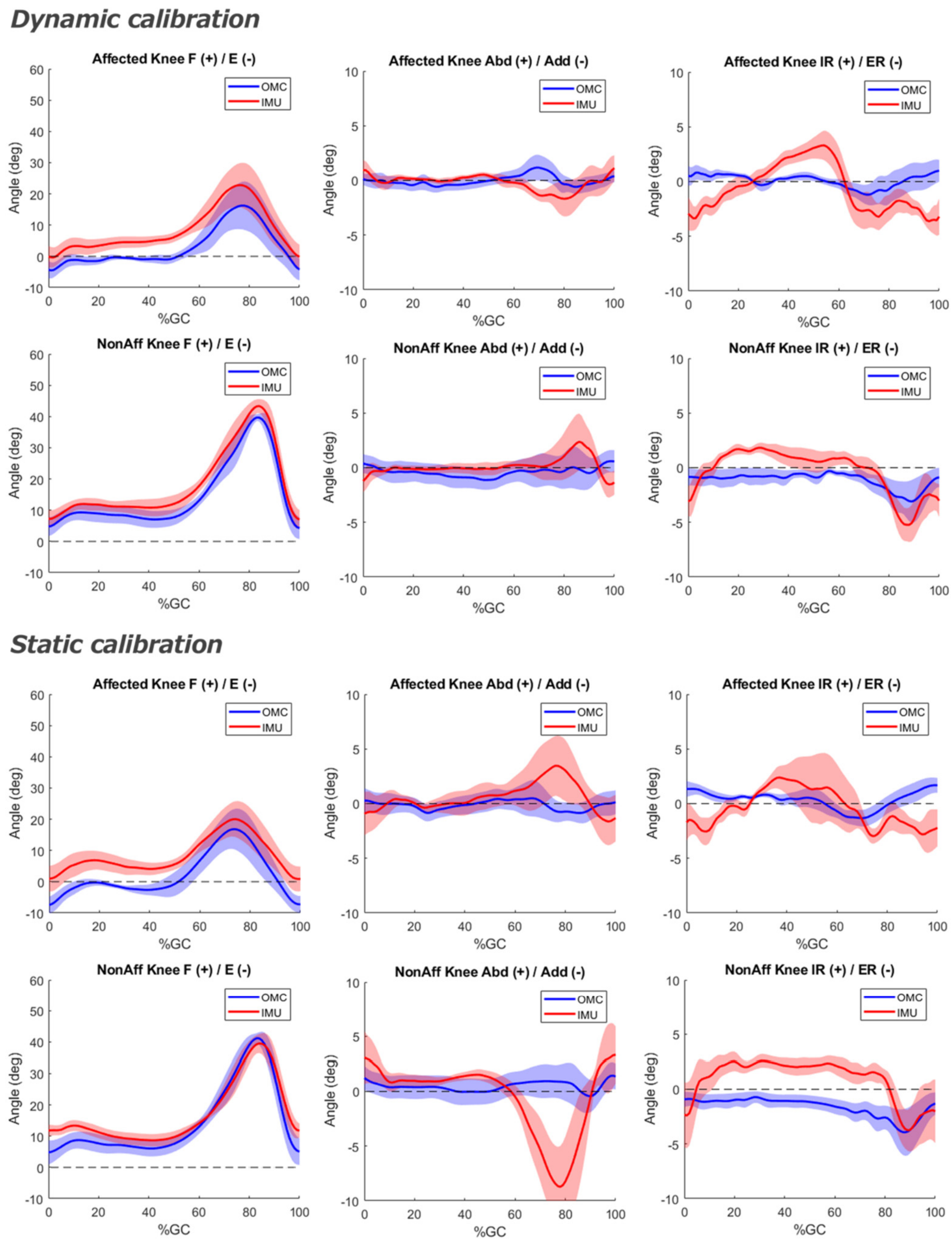

Fig. 3. Knee angles for the dynamic and static calibrations. Abd/Add: abduction/adduction, F/E: flexion/extension, IMU: inertial measurement unit, IR/ER: internal/external rotation, NonAff: non affected, OMC: optical motion capture.

Our RMSE results are lower than the knee's MCID on both sides for the dynamic calibration. The ROMd that we obtained with the dynamic calibration is also lower than the reported knee's MCID. Thus, it would be possible to opt for the dynamic calibration to monitor patient's improvements regarding knee ROM. The same research group also reported MCID values for the hip sagittal ROM: 5.81° (affected side) and 2.86° (unaffected side) [42]. Our RMSE values are smaller than the hip's MCID only for the affected side using the dynamic calibration. However, the ROMd that we measured is lower than the hip's MCID for the unaffected side during the dynamic calibration. It is important to point out that Guzik et al. [41,42] determined MCID in chronic patients that walked unassisted with a walking speed above 0.4 m.s^{-1}. Therefore, we can anticipate that these values would be higher in the acute phase., Improvements in either data collection process or IMU algorithms must be made to monitor hip ROM improvements. Largest progresses being made in the first three months [43,44], we believe that Xsens' IMU could be used to monitor joint ROM improvements.

In this study, we evaluated Xsens MVN accuracy in a laboratory setting, where the ferromagnetic environment is known and can be controlled. This will be more difficult to control in ambulatory settings [45]. To limit inaccuracies in orientation, the latter authors recommend calibrating IMU in the environment where the data collection will be performed. Therefore, if the subject is expected to walk next to a ferromagnetic environment that cannot be removed, IMUs should be calibrated like this. Thus, Xsens' accuracy might be lower than what we demonstrated in this study.

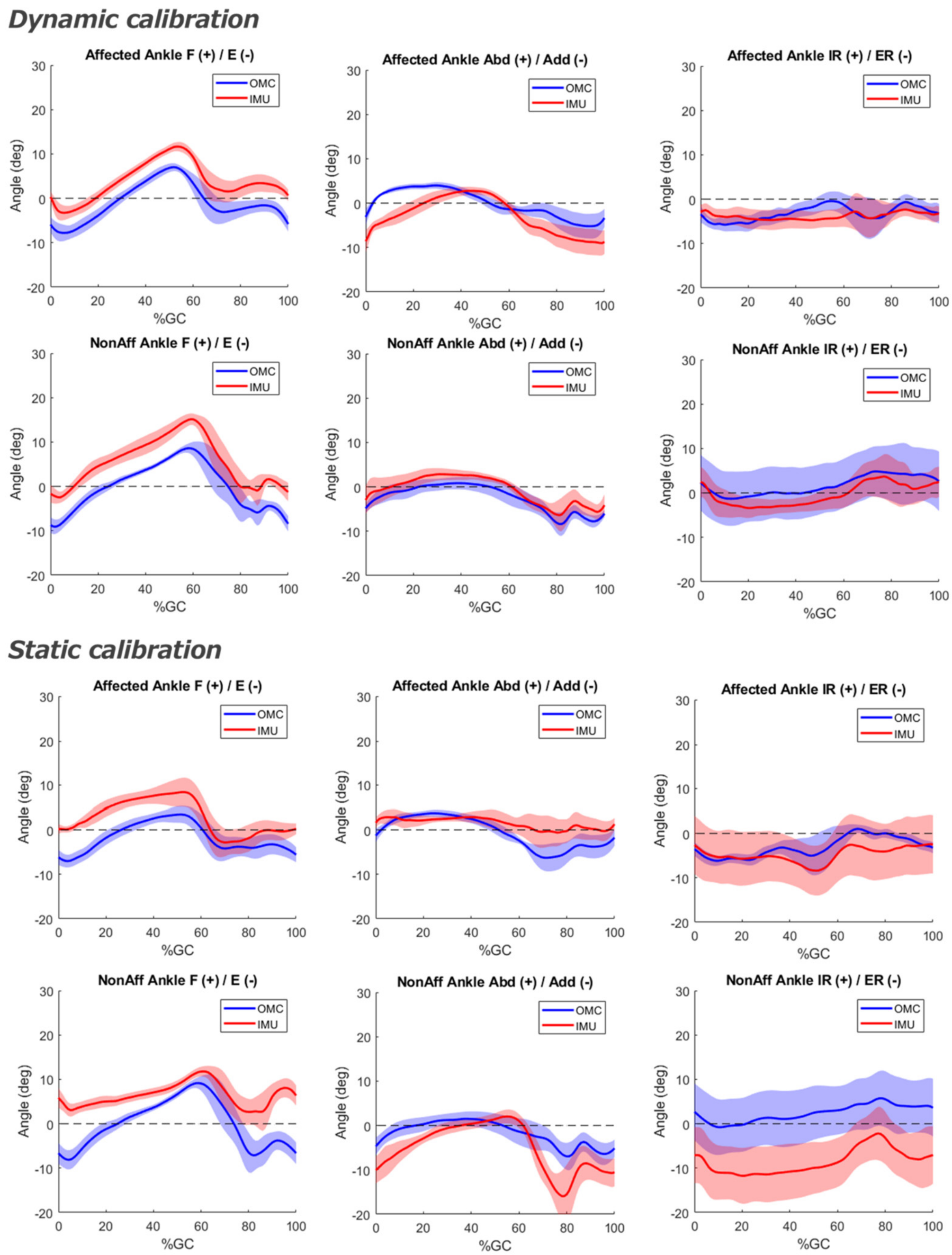

Fig. 4. Ankle angles for the dynamic and static calibrations. Abd/Add: abduction/adduction, F/E: flexion/extension, IMU: inertial measurement unit, IR/ER: internal/external rotation, NonAff: non affected, OMC: optical motion capture.

Several other limitations can be pointed out. Walking ability varied among patients, from unassisted walking to tripod canes and orthoses. While this is interesting in testing Xsens' capabilities, a more uniform group would have eased the validation. Therefore, a greater cohort could enable a more comprehensive validation of this system and calibration procedure. We also did not evaluate whether Xsens' events identification was accurate. While this was not a problem for most of our subjects, some additional foot contacts were identified, particularly in patients that had a small foot clearance.

5. Conclusion

In this specific population, the calibration type, and slow walking speed, does not seem to impact Xsens' accuracy to a great extent. Nevertheless, dynamic calibration provides slightly better results, especially for A/A and F/E, as functional axes are better defined. Considering these results, and the patient's walking ability, we recommend using this calibration instead of the static one.

In perspectives, the calibration method could be modified by manually increasing the calibration time for slow walker, or by adding new movements to improve functional axes' definition. Long term goals will be to promote IMU's usability in clinical settings through a routine use during rehabilitation and a feedback approach that may use gamification.

CRediT authorship contribution statement

Ariane P. Lallès: Writing – original draft, Investigation, Formal analysis, Data curation, Conceptualization. **Geoffroy Moucheboeuf:**

Table 2
Range of movement values obtained with both systems during the dynamic calibration for the post-stroke group (mean ± SD). All results are expressed in degrees.

			OMC-dynamic		IMU-dynamic	
			Affected	Non-afFected	Affected	Non-affected
ROM	Hip	F/E	26.3 ± 3.6	26.3 ± 5.6	20.3 ± 6.6	25.0 ± 4.5
		A/A	9.3 ± 4.6	7.8 ± 1.4	11.6 ± 4.6	10.1 ± 2.5
		I/E	12.5 ± 3.3	11.4 ± 4.6	9.2 ± 2.7	9.2 ± 1.0
	Knee	F/E	27.2 ± 11.0	43.2 ± 5.3	28.9 ± 11.3	45.1 ± 7.1
		A/A	4.4 ± 1.7	4.8 ± 1.4	6.6 ± 2.4	9.2 ± 3.8
		I/E	4.9 ± 1.7	5.1 ± 4.5	11.9 ± 4.5	10.4 ± 3.9
	Ankle	F/E	17.4 ± 5.9	25.0 ± 7.8	17.9 ± 6.1	26.4 ± 8.0
		A/A	12.1 ± 6.4	14.3 ± 4.7	14.6 ± 7.6	14.1 ± 5.3
		I/E	16.3 ± 6.8	13.7 ± 6.0	12.5 ± 8.8	13.6 ± 7.1

A/A: abduction/adduction, F/E: flexion/extension, I/E: internal/external rotation, IMU: inertial measurement unit, OMC: optical motion capture, ROM: range of motion.

Table 3
Errors between systems for the post-stroke group (mean ± SD). All results are expressed in degrees. Negative values imply an overestimation of the Xsens system compared to the optical motion capture. Errors below 5° are reported in green, between 5.01° and 9.99° in orange, and above 10° in red.

			Dynamic calibration		Static calibration	
			Affected	Non-affected	Affected	Non-affected
Bias	**Hip**	F/E	-3.7 ± 1.0	-2.9 ± 2.2	-4.0 ± 4.0	-2.9 ± 5.4
		A/A	-3.1 ± 2.6	1.0 ± 3.9	-2.3 ± 4.3	-0.9 ± 3.9
		I/E‡‡	0.3 ± 3.7	3.7 ± 14.9	2.1 ± 12.4	0.0 ± 16.5
	Knee	F/E	-5.4 ± 5.9	-3.5 ± 5.0	-6.2 ± 7.5	-2.3 ± 5.4
		A/A	0.2 ± 1.0	-0.5 ± 2.7	-0.7 ± 1.4	1.0 ± 2.8
		I/E‡‡	0.7 ± 1.3	-0.8 ± 2.2	0.8 ± 1.2	-2.5 ± 2.4
	Ankle	F/E	-4.7 ± 2.9	-5.6 ± 2.7	-4.6 ± 4.2	-6.5 ± 4.2
		A/A*	2.8 ± 2.5	-1.8 ± 4.1	-2.1 ± 1.9	3.2 ± 7.0
		I/E	0.6 ± 3.4	2.1 ± 14.9	1.7 ± 16.1	10.9 ± 18.5
TEM	**Hip**	F/E	2.7 ± 2.1	1.4 ± 0.4	2.1 ± 0.9	2.2 ± 1.3
		A/A	4.1 ± 3.2	2.5 ± 1.2	3.4 ± 2.5	5.3 ± 2.8
		I/E	2.7 ± 1.1	2.5 ± 0.5	2.9 ± 1.1	2.4 ± 0.5
	Knee	F/E††	1.8 ± 0.8	1.6 ± 0.5	2.9 ± 1.0	3.6 ± 1.7
		A/A††	1.7 ± 1.1	2.7 ± 1.2	3.3 ± 1.7	5.4 ± 2.8
		I/E	3.4 ± 1.5	2.4 ± 0.9	3.6 ± 1.3	2.8 ± 2.0
	Ankle	F/E††	1.6 ± 0.5	1.8 ± 0.6	2.4 ± 1.2	4.5 ± 2.6
		A/A†	3.0 ± 1.0	2.5 ± 1.7	4.0 ± 0.7	5.2 ± 2.8
		I/E	2.4 ± 0.6	2.5 ± 0.8	2.6 ± 0.8	2.9 ± 1.8
RMSE	**Hip**	F/E	4.8 ± 1.7	3.6 ± 1.3	5.1 ± 3.1	5.5 ± 3.4
		A/A	5.5 ± 3.6	4.4 ± 1.6	5.3 ± 3.4	6.5 ± 2.7
		I/E	3.9 ± 2.3	9.5 ± 11.8	10.5 ± 6.1	13.6 ± 7.6
	Knee	F/E	6.5 ± 4.8	5.2 ± 3.2	8.4 ± 5.3	6.5 ± 1.9
		A/A†‡	1.9 ± 1.1	3.5 ± 1.9	3.6 ± 1.6	6.1 ± 2.7
		I/E	3.7 ± 1.5	3.1 ± 1.3	3.9 ± 1.1	4.3 ± 2.0
	Ankle	F/E	5.0 ± 2.8	5.9 ± 2.7	5.9 ± 3.1	8.0 ± 4.6
		A/A*†	4.5 ± 1.7	4.3 ± 3.0	4.8 ± 0.9	8.7 ± 3.1
		I/E	3.7 ± 1.8	11.2 ± 9.1	12.0 ± 9.9	15.0 ± 15.2
ROMd	**Hip**	F/E	6.1 ± 5.2	1.3 ± 2.1	4.6 ± 2.2	4.6 ± 4.2
		A/A	-2.3 ± 7.3	-2.3 ± 2.5	-4.6 ± 6.2	-8.5 ± 7.5
		I/E	3.2 ± 3.5	2.2 ± 5.1	2.0 ± 4.5	1.3 ± 4.4
	Knee	F/E†††	-1.7 ± 1.5	-1.8 ± 3.6	7.5 ± 4.9	7.2 ± 10.1
		A/A††	-2.2 ± 2.4	-4.4 ± 4.8	-8.3 ± 6.2	-15.1 ± 8.4
		I/E	-7.0 ± 3.0	-5.3 ± 4.6	-8.0 ± 2.3	-6.3 ± 4.9
	Ankle	F/E*†	-0.5 ± 1.7	-1.4 ± 0.6	-1.4 ± 3.2	7.1 ± 8.0
		A/A*	-2.5 ± 3.3	0.2 ± 2.3	-1.3 ± 6.2	-11.0 ± 11.6
		I/E	3.8 ± 6.1	0.1 ± 6.1	3.9 ± 3.6	-2.0 ± 7.9

A/A: abduction/adduction, F/E: flexion/ extension, I/E: internal/external rotation, RMSE: root mean square error, ROMd: range of motion differences, TEM: typical error of measurement, *: significant interaction between the calibration type and side (*: p≤0.05, **: p≤0.01, ***: p≤0.001), †: significant effect on the calibration type, ‡: significant effect on the side.

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Author contributions

All authors attest that they meet the current International Committee of Medical Journal Editors (ICMJE) criteria for Authorship.

Informed consent and patient details

The authors declare that they obtained a written informed consent from the patients and/or volunteers included in the article and that this report does not contain any personal information that could lead to their identification.

Human and animal rights

The authors declare that the work described has been carried out in accordance with the Declaration of Helsinki of the World Medical Association revised in 2013 for experiments involving humans as well as in accordance with the EU Directive 2010/63/EU for animal experiments.

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Declaration of competing interest

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